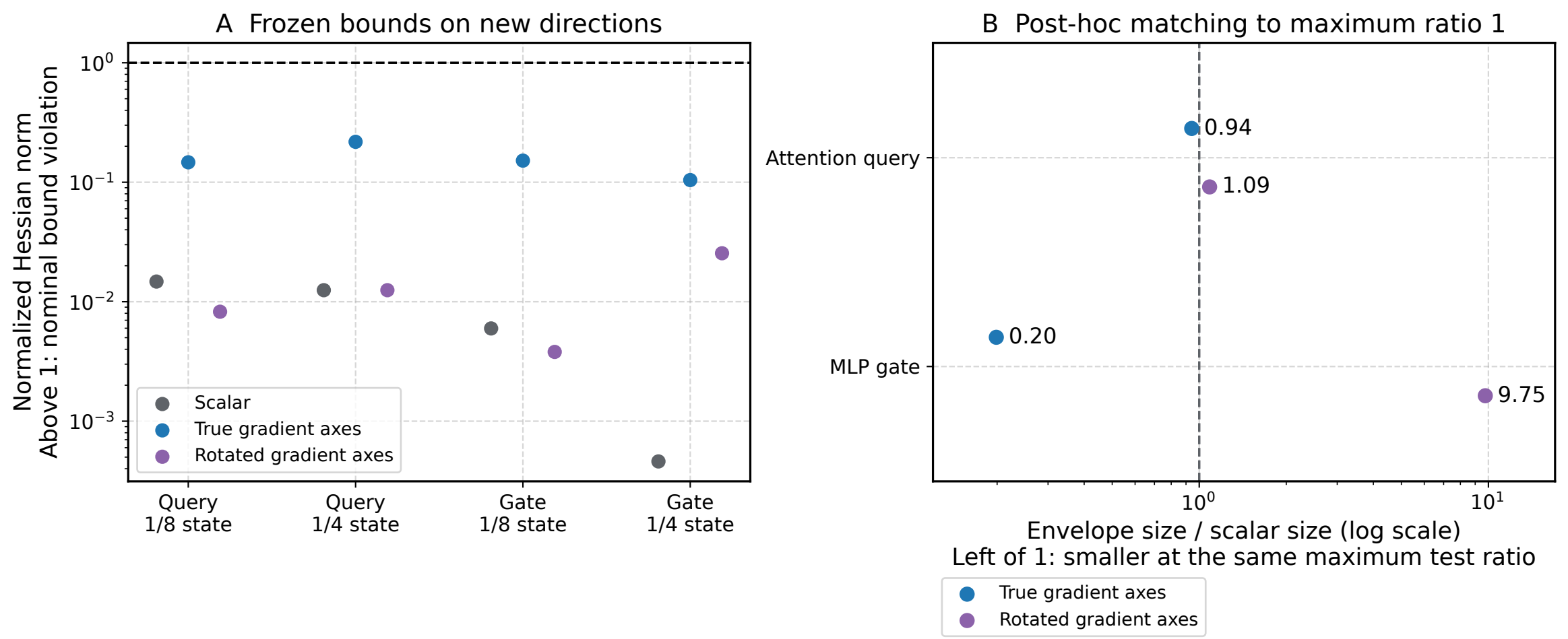
A Frozen bounds on new directions
Normalized Hessian norm
Above 1: nominal bound violation
10^{0}
10^{-1}
10^{-2}
10^{-3}
Scalar
True gradient axes
Rotated gradient axes
Query 1/8 state
Query 1/4 state
Gate 1/8 state
Gate 1/4 state
B Post-hoc matching to maximum ratio 1
Attention query
MLP gate
0.94
1.09
0.20
9.75
10^{0}
10^{1}
Envelope size / scalar size (log scale)
Left of 1: smaller at the same maximum test ratio
True gradient axes
Rotated gradient axes